

183. Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-FR4 and 800GBASE-LR4

183.1 Overview

This clause specifies the 800GBASE-FR4 and 800GBASE-LR4 Physical Medium Dependent (PMD) types together with the single-mode fiber medium. The optical signals generated by these two PMD types are modulated using a 4-level pulse amplitude modulation (PAM4) format. The PMDs provide point-to-point 800 Gb/s Ethernet links over four wavelength division multiplexing (WDM) lanes on single-mode fiber, with a reach of up to at least 2 km or 10 km, respectively.

When forming a complete Physical Layer, a PMD shall be connected to the appropriate PMA as shown in Table 183–1 to the medium through the MDI.

Table 183–1—Physical Layer clauses associated with the 800GBASE-FR4 and 800GBASE-LR4 PMDs

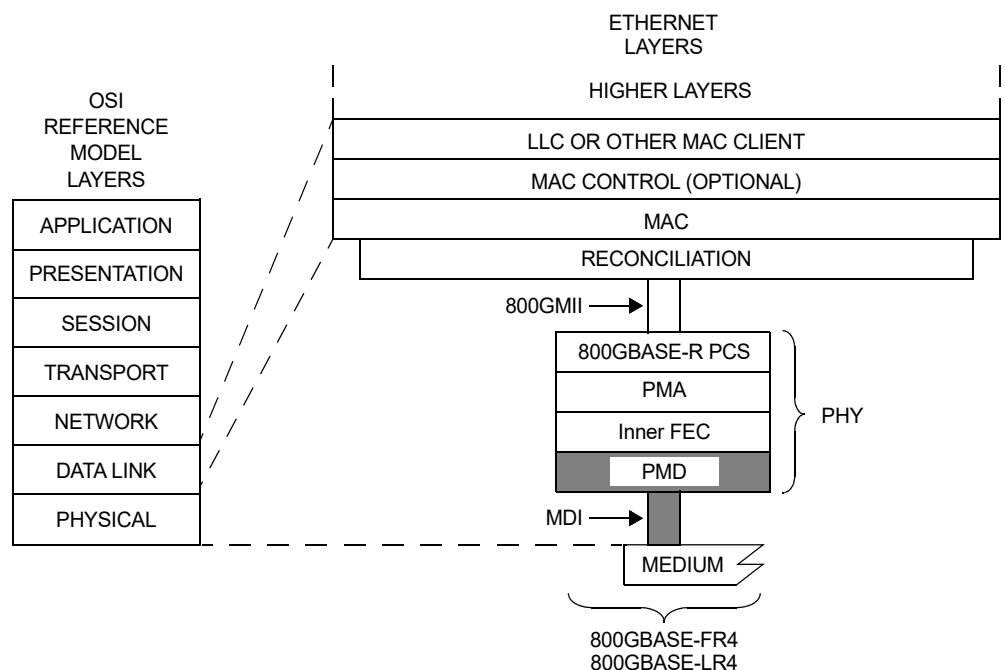
Associated clause	800GBASE-FR4 800GBASE-LR4
90—Time Synchronization	Optional
120F—800GAUI-8 C2C	Optional ^a
120G—800GAUI-8 C2M	Optional ^a
170—800 Gb/s RS	Required
170—800GMII ^b	Optional
171—800GMII Extender	Optional
172—800GBASE-R PCS	Required
173—800GBASE-R BM-PMA	Conditional ^c
176—800GBASE-R SM-PMA	Required ^c
176C—800GAUI-4 C2C	Optional ^a
176D—800GAUI-4 C2M	Optional ^a
177—800GBASE-R Inner FEC	Required

^a One or two 800GAUI-n may be instantiated within a 800GBASE-FR4 PHY or 800GBASE-LR4 PHY as described in 176B.7.1.

^b The 800GMII is an optional interface. However, if the 800GMII is not implemented, a conforming implementation behaves functionally as though the RS and 800GMII were present.

^c If one or two 800GAUI-n are implemented in a PHY, additional 800GBASE-R BM-PMA or SM-PMA sublayers are required according to 176B.7.1.

Figure 183–1 shows the relationship of the PMD and MDI (shown shaded) along with other Ethernet sublayers to the ISO/IEC Open System Interconnection (OSI) reference model. 800 Gb/s Ethernet is introduced in Clause 169 and the purpose of each PHY sublayer is summarized in 169.2.



800GMII = 800 Gb/s MEDIA INDEPENDENT INTERFACE
FEC = FORWARD ERROR CORRECTION
LLC = LOGICAL LINK CONTROL
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
PCS = PHYSICAL CODING SUBLAYER

PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT
FR4 = PMD FOR SINGLE-MODE FIBER — 2 km
LR4 = PMD FOR SINGLE-MODE FIBER — 10 km

Figure 183–1—800GBASE-FR4 and 800GBASE-LR4 PMDs relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

183.2 Error ratio allocation

A complete PHY shall meet the specifications in 174A.6.

With a compliant input signal, a PMD receiver is expected to meet the block error ratio (BLER) of 1.45×10^{-11} (see 174A.6), measured at the PMA nearest to the PMD using the method described in 174A.9, with BER_{added} equal to 6.4×10^{-5} .

With a compliant input signal, a PHY receiver is expected to meet the BLER of 1.45×10^{-11} (see 174A.6), measured at the PCS using the method described in 174A.11, with BER_{added} equal to 3.2×10^{-5} .

183.3 PMD service interface

This subclause specifies the services provided by the PMD. The service interface for this PMD is described in an abstract manner and does not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMD and the PMD client (Inner FEC immediately above the PMD). The PMD translates the encoded data to and from signals suitable for the specified medium.

The PMD service interface is an instance of the inter-sublayer service interface defined in 169.3. For PMD:IS_UNITDATA_*i*.request primitive in the transmit direction and PMD:IS_UNITDATA_*i*.indication primitive in the receive direction, where *i* = 0 to *n*−1, the tx_symbol and rx_symbol parameters are parallel PAM4 symbol streams with a nominal signaling rate of 113.4375 GBd. The number of parallel streams, *n*, is 4. Each instance of tx_symbol and rx_symbol takes on one of four (PAM4) values: 0, 1, 2, or 3. In order to support the Inner FEC soft-decision decoder (see 177.5.5), rx_symbol may take on an implementation-dependent set of values beyond the scope of this standard.

The SIGNAL_OK parameter of the PMD:IS_SIGNAL.indication primitive (see 116.3.3.3) corresponds to the variable rts_status of the ready-to-send (RTS) function (see 183.5.12), as defined in 178B.8.2.1. When SIGNAL_OK is either IN_PROGRESS or FAIL, the rx_symbol parameters of PMD:IS_UNITDATA_*i*.indication primitive on all lanes are unspecified.

The PMD receives signal status information from the PMD client using the PMD:IS_SIGNAL.request primitive (see 116.3.3.4).

183.4 Delay and Skew

183.4.1 Delay constraints

The sum of the transmit and receive delays at one end of the link contributed by the 800GBASE-FR4 or 800GBASE-LR4 PMD including 2 m of fiber in one direction shall be no more than 59392 bit times (116 pause_quanta or 74.24 ns). A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 169.4 and its references.

183.4.2 Skew constraints

The Skew (relative delay between the lanes) and Skew Variation are kept within limits so that the information on the lanes can be reassembled by the PCS. Skew and Skew Variation are defined in 169.5 and specified at the points SP1 to SP6 shown in Figure 169–4 and Figure 169–5.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 25 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 36 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 116 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 127 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 169.5. The measurements of Skew and Skew Variation are defined in 87.8.2.

183.5 PMD functional specifications

The 800GBASE-FR4 and 800GBASE-LR4 PMDs perform the Transmit and Receive functions, which convey data between the PMD service interface and the MDI.

183.5.1 Block diagram

The PMD block is shown in Figure 183–2. For purposes of system conformance, the PMD sublayer is standardized at the points described in this subclause. The optical transmit signal is defined at the output end of a single-mode fiber patch cord (TP2), between 2 m and 5 m in length. Unless specified otherwise, all transmitter measurements and tests defined in 183.9 are made at TP2. The optical receive signal is defined at the output of the fiber optic cabling (TP3) at the MDI (see 183.8.3). Unless specified otherwise, all receiver measurements and tests defined in 183.9 are made at TP3.

While TP2 and the patch cord used for transmitter measurements and tests are shown in Figure 183–2, they are not part of the medium. The transmit MDI is connected to the medium for use, or to the patch cord and TP2 for transmitter measurements and tests.

The inter-sublayer link training (ILT) and RTS functions indicated in Figure 183–2 are defined in Annex 178B.

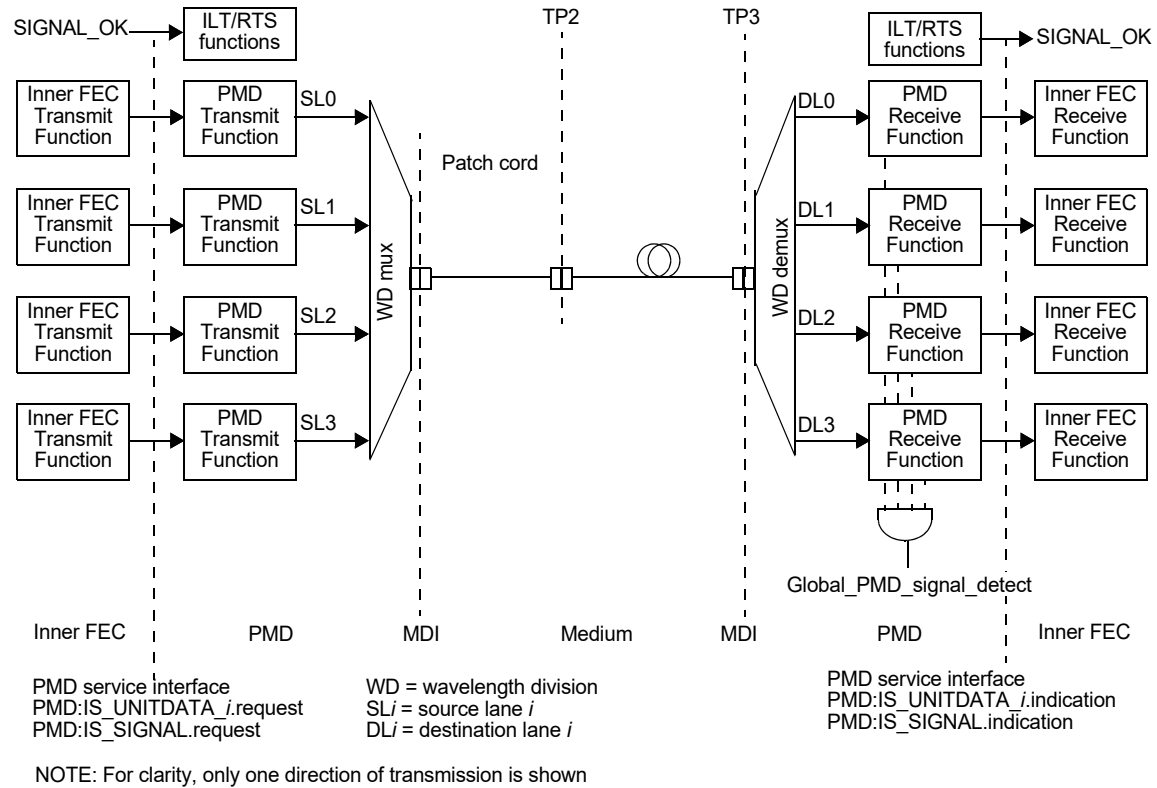


Figure 183–2—Block diagram for 800GBASE-FR4 and 800GBASE-LR4 transmit/receive paths

183.5.2 PMD transmit function

The PMD transmit function has two operating mode: DATA and TRAINING. The operating mode is controlled by the tx_mode variable of the ILT function (see 183.5.12); it is DATA when tx_mode = DATA, and TRAINING otherwise.

When operating in DATA mode, the PMD Transmit function shall convert the four symbol streams requested by the PMD service interface `PMD:IS_UNITDATA_0.request` primitive to `PMD:IS_UNITDATA_3.request` primitive into four separate optical signals. The four optical signals shall then be wavelength division multiplexed and delivered to the MDI, all according to the transmit optical specifications in this clause.

The polarity of the PMD output on each of the lanes is either normal or inverted. When normal, the four optical power levels at the MDI output, in order from lowest to highest, correspond to `tx_symbol` values zero, one, two, and three, respectively. When inverted, the four optical power levels at the MDI output, in order from highest to lowest, correspond to `tx_symbol` values zero, one, two, and three, respectively.

If training is enabled, either normal or inverted output polarity may be used, since the PMD receiver in the peer interface detects the polarity and corrects it if necessary (see 178B.7.7). If training is disabled, the output polarity of each lane shall be normal. Output polarity may be controlled by management in an implementation dependent manner.

When operating in TRAINING mode, the PAM4 symbol stream on each lane is provided by the ILT function (see 178B.7).

183.5.3 PMD receive function

The PMD Receive function shall demultiplex the composite optical signal received from the MDI into four separate optical signals. The four optical signals shall then be converted into four symbol streams for delivery to the PMD service interface using the `PMD:IS_UNITDATA_0.indication` primitive to `PMD:IS_UNITDATA_3.indication` primitive, all according to the receive optical specifications in this clause.

The polarity of the PMD input on each of the lanes is either normal or inverted. When normal, the four optical power levels at the MDI input, in order from lowest to highest, correspond to `tx_symbol` values zero, one, two, and three, respectively. When inverted, the four optical power levels at the MDI input, in order from highest to lowest, correspond to `tx_symbol` values zero, one, two, and three, respectively.

If training is enabled, the PMD shall detect the polarity during training and correct it if necessary (see 178B.7.7). If training is disabled, the input polarity of each lane shall be normal. Input polarity may be controlled by management in an implementation dependent manner.

183.5.4 PMD global signal detect function

The variable `Global_PMD_signal_detect` is a global indicator of the presence of optical signals on all n lanes and it is set to the logical-and of `PMD_signal_detecti` from all of the PMD lanes.

183.5.5 PMD lane-by-lane signal detect function

The PMD lane-by-lane signal detect function is used by the PMD to indicate sufficient optical power is detected at the receiver input on each lane. For each lane i , where i is in the range 0 to $n-1$, a receiver shall set the variable `PMD_signal_detecti` to either 0 or 1 according to the criteria in Table 183–2

Various implementations of the signal detect function are permitted by this standard. This standard imposes no response time requirements on the generation of the `PMD_signal_detecti` variable.

Implementations need to provide adequate margin between the input optical power level at which the `PMD_signal_detecti` variable is set to 1 and the inherent noise level of the PMD, including the effects of crosstalk, power supply noise, etc.

Table 183–2—Signal detection criteria for each lane

Receive conditions	PMD_signal_detect_ <i>i</i>
Average optical power at TP3 ≤ -16 dBm	0
For all lanes; [The input signal is a compliant 800GBASE-R signal input] ^a AND [Optical power at TP3 $\geq$ average receive power, each lane (min) in Table 183–7]	1
All other conditions	Unspecified, may be 0 or 1

^a This is a condition for testing of the receiver, not a requirement for the receiver to detect that the signal is valid.

183.5.6 PMD reset function

If the variable PMD_reset is asserted, the PMD shall be reset as defined in 45.2.1.1.1.

PMD reset shall reset the ILT and RTS functions associated with the PMD (see 183.5.12).

183.5.7 PMD global transmit disable function (optional)

The PMD global transmit disable function is optional and allows all of the optical transmitters to be disabled.

When the Global_PMD_transmit_disable variable is set to one, this function shall turn off all of the optical transmitters so that each transmitter meets the requirements of the average launch power of the OFF transmitter in Table 183–6.

183.5.8 PMD lane-by-lane transmit disable function

The PMD lane-by-lane transmit disable function allows the optical transmitters in each lane to be selectively disabled. This function is used by the ILT function. It may optionally be provided through control variables PMD_transmit_disable_ *i*.

If either PMD_transmit_disable_ *i* or the variable tx_disable in lane *i* of the ILT function (see 178B.8.2.1) is set to one (where *i* represents the lane number in the range 0:*n*–1), this function shall turn off the associated optical transmitter so that the transmitter meets the requirements of the average launch power of the OFF transmitter in Table 183–6.

If a PMD_fault is detected, then the PMD may set each PMD_transmit_disable_ *i* variable to one, turning off the optical transmitter in each lane.

183.5.9 PMD fault function (optional)

If the PMD has detected a local fault on any of the transmit or receive paths, the PMD shall set PMD_fault to one.

183.5.10 PMD transmit fault function (optional)

If the PMD has detected a local fault on any transmit lane, the PMD shall set the PMD_transmit_fault variable to one.

183.5.11 PMD receive fault function (optional)

If the PMD has detected a local fault on any receive lane, the PMD shall set the PMD_receive_fault variable to one.

183.5.12 Autonomous path startup (APSU) functions

A PMD shall provide the RTS function and the ILT function with a training frame of O1 format, specified in Annex 178B.

When the variable mr_training_enable is true, the ILT function is used to request changes to the peer transmitter state (modulation, training pattern, and precoder state), indicate the receiver state, and coordinate the transition to DATA mode.

When tx_mode = LOCAL_PATTERN (see 178B.8.3.1), the PMD transmits PRBS31 encoded by Inner FEC (see 177.6.1.1).

The default value of max_wait_timer_duration is 60, corresponding to a duration of 60 seconds for max_wait_timer (see 178B.8.3.3).

183.6 Wavelength-division-multiplexed lane assignments

The wavelength range for each lane of the 800GBASE-FR4 is defined in Table 183–3. The 800GBASE-FR4 center wavelengths are members of the CWDM wavelength grid defined in ITU-T G.694.2 and are spaced at 20 nm.

The wavelength range for each lane of the 800GBASE-LR4 is defined in Table 183–4. The 800GBASE-LR4 center frequencies are members of the frequency grid for 100 GHz spacing defined in ITU-T G.694.1 and are spaced at 800 GHz.

Table 183–3—800GBASE-FR4 wavelength-division-multiplexed lane assignments

Lane	Center wavelength	Wavelength range
0	1271 nm	1264.5 to 1277.5 nm
1	1291 nm	1284.5 to 1297.5 nm
2	1311 nm	1304.5 to 1317.5 nm
3	1331 nm	1324.5 to 1337.5 nm

Table 183–4—800GBASE-LR4 wavelength-division-multiplexed lane assignments

Lane	Center frequency	Center wavelength	Wavelength range
0	231.4 THz	1295.56 nm	1294.53 to 1296.59 nm
1	230.6 THz	1300.05 nm	1299.02 to 1301.09 nm
2	229.8 THz	1304.58 nm	1303.54 to 1305.63 nm
3	229 THz	1309.14 nm	1308.09 to 1310.19 nm

NOTE—Each functional lane, denoted SL_i at the transmitter and DL_i at the receiver, is mapped to a specific wavelength to support ILT operation.

183.7 PMD to MDI optical specifications

The operating ranges for the PMDs are defined in Table 183–5. A 800GBASE-FR4 or 800GBASE-LR4 compliant PMD operates on type G.652.D, G.657.A1, or G.657.A2 single-mode fibers. A PMD that exceeds the operating range requirement while meeting all other optical specifications is considered compliant (e.g., a 800GBASE-FR4 PMD operating at 2.5 km meets the operating range requirement of 2 m to 2 km).

Table 183–5—800GBASE-FR4 and 800GBASE-LR4 operating ranges

PMD type	Required operating range
800GBASE-FR4	2 m to 2 km
800GBASE-LR4	2 m to 10 km

183.7.1 Transmitter optical specifications

The transmitter shall meet the specifications defined in Table 183–6 per the definitions in 183.9.

Table 183–6—800GBASE-FR4 and 800GBASE-LR4 transmit characteristics

Description	Defined in	800GBASE-FR4	800GBASE-LR4	Unit
Signaling rate, each lane (range)	—	113.4375 ± 50 ppm		GBd
Modulation format	—	PAM4		—
Lane wavelengths (range)	183.9.3	1264.5 to 1277.5 1284.5 to 1297.5 1304.5 to 1317.5 1324.5 to 1337.5	1294.53 to 1296.59 1299.02 to 1301.09 1303.54 to 1305.63 1308.09 to 1310.19	nm
Side-mode suppression ratio (SMSR), each lane (min)	183.9.3	30		dB
Total average launch power (max)	183.9.4	10.9	11.5	dBm
Average launch power, each lane (max)	183.9.4	4.9	5.5	dBm
Average launch power, each lane ^a (min)	183.9.4	−2.2 ^b	−1.1 ^c	dBm
Outer optical modulation amplitude (OMA _{outer}), each lane (max)	183.9.5	4.8	5.7	dBm
Outer optical modulation amplitude (OMA _{outer}), each lane (min) for max(TECQ, TDECQ) < 0.9 dB for 0.9 dB ≤ max(TECQ, TDECQ) ≤ 3.4 dB for max(TECQ, TDECQ) < 1.4 dB for 1.4 dB ≤ max(TECQ, TDECQ) ≤ 3.9 dB	183.9.5	0.8 Equation (183–1) — —	— — 1.9 Equation (183–2)	dBm dBm dBm dBm
Difference in launch power between any two lanes (OMA _{outer}) (max)	183.9.5	4	3	dB
Transmitter and dispersion eye closure for PAM4 (TDECQ), each lane (max)	183.9.6	3.4	3.9	dB
Transmitter eye closure for PAM4 (TECQ), each lane (max)	183.9.7	3.4	3.2	dB
TDECQ – TECQ , each lane (max)	—	2.5	2.5	dB
Transmitter functional symbol error histogram, each lane (max)	183.9.8	See Table 180–17		—
Transmitter overshoot and undershoot, each lane (max)	183.9.9	27		%
Transmitter power excursion, each lane (max)	183.9.10	3.5	4.4	dBm
Extinction ratio, each lane (min)	183.9.11	3.5		dB
Transmitter transition time, each lane (max)	183.9.12	8	13	ps
Average launch power of OFF transmitter, each lane (max)	—	−16	−16	dBm
RIN _x OMA, each lane (max)	183.9.13	−139		dB/Hz
Optical return loss tolerance, each lane (max)	183.9.13	17.1	15.6	dB
Transmitter reflectance ^d , each lane (max)	—	−26		dB

^a Average launch power, each lane (min) is not the principal indicator of signal strength. A transmitter with launch power below this value cannot be compliant; however, a value above this does not ensure compliance.

^b Average launch power of −2.2 dBm corresponds to an OMA of 0.8 dBm with an infinite extinction ratio.

^c Average launch power of −1.1 dBm corresponds to an OMA of 1.9 dBm with an infinite extinction ratio.

^dTransmitter reflectance is defined looking into the transmitter.

$$-0.1 + \max(\text{TECQ}, \text{TDECQ}) \tag{183-1}$$

$$0.5 + \max(\text{TECQ}, \text{TDECQ}) \tag{183-2}$$

The values for $\text{OMA}_{\text{outer}}$ each lane (min) in Table 183–6 vary with $\max(\text{TECQ}, \text{TDECQ})$. The relationships are illustrated in Figure 183–3 along with the values for $\text{OMA}_{\text{outer}}$ each lane (max).

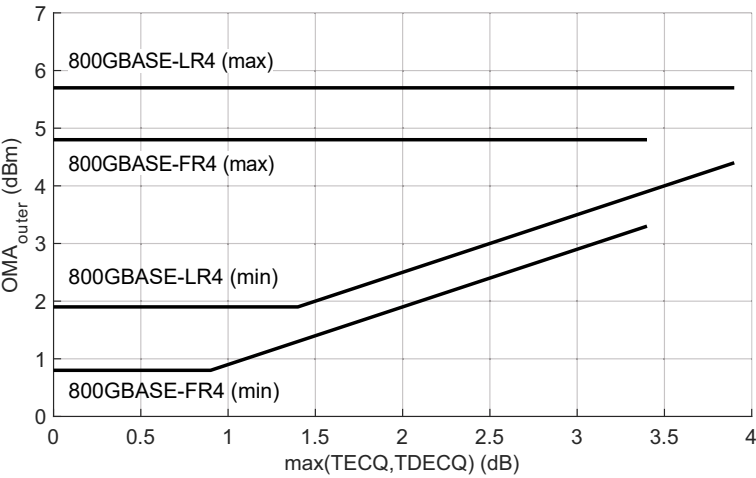


Figure 183–3— $\text{OMA}_{\text{outer}}$ each lane (max) and $\text{OMA}_{\text{outer}}$ each lane (min) versus $\max(\text{TECQ}, \text{TDECQ})$ for 800GBASE-FR4 and 800GBASE-LR4

183.7.2 800GBASE-FR4 and 800GBASE-LR4 receive optical specifications

The receiver shall meet the specifications in Table 183–7 per the definitions in 183.9.

Table 183–7—800GBASE-FR4 and 800GBASE-LR4 receive characteristics

Description	Defined in	800GBASE-FR4	800GBASE-LR4	Unit
Signaling rate, each lane (range)	—	113.4375 ± 50 ppm		GBd
Modulation format	—	PAM4		—
Lane wavelengths (range)	183.9.3	1264.5 to 1277.5 1284.5 to 1297.5 1304.5 to 1317.5 1324.5 to 1337.5	1294.53 to 1296.59 1299.02 to 1301.09 1303.54 to 1305.63 1308.09 to 1310.19	nm
Damage threshold ^a , each lane	—	5.9	6.5	dBm
Average receive power, each lane (max)	—	4.9	5.5	dBm
Average receive power, each lane ^b (min)	—	−6.2	−7.4	dBm
Receive power (OMA _{outer}), each lane (max)	—	4.8	5.7	dBm
Difference in receive power between any two lanes (OMA _{outer}) (max)	—	4.1	3.3	dB
Receiver reflectance (max)	—	−26		dB
Receiver sensitivity (OMA _{outer}), each lane ^c (max) for TECQ < 0.9 dB for 0.9 dB ≤ TECQ ≤ SECQ for TECQ < 1.4 dB for 1.4 dB ≤ TECQ ≤ SECQ	183.9.14	−3.7 −4.6 + TECQ — —	— — −5.5 −6.9 + TECQ	dBm dBm dBm dBm
Stressed receiver sensitivity (OMA _{outer}), each lane ^c (max)	183.9.15	−1.2	−3	dBm
Conditions of stressed receiver sensitivity test: ^d				
Stressed eye closure for PAM4 (SECQ), lane under test	—	3.4	3.9	dB
OMA _{outer} of each aggressor lane	—	0.8	0.3	dBm

^a The receiver shall be able to tolerate, without damage, continuous exposure to an optical input signal having this average power level.

^b Average receive power, each lane (min) is not the principal indicator of signal strength. A received power below this value cannot be compliant; however, a value above this does not ensure compliance.

^c Measured with conformance test signal at TP3 (see 183.8) for the BLER specified in 183.2.

^d These test conditions are for measuring stressed receiver sensitivity. They are not characteristics of the receiver.

The values for receiver sensitivity (OMA_{outer}), each lane (max) in Table 183–7 vary depending on the value of TECQ for the test source used for measurement. Figure 183–4 illustrates this.

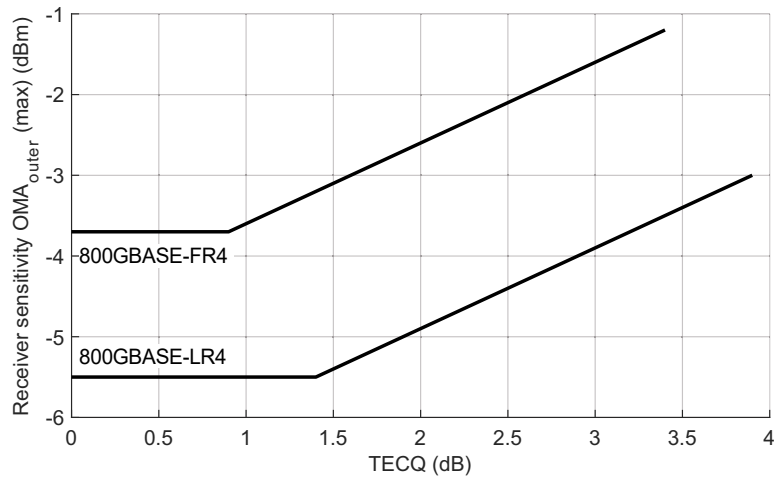


Figure 183-4—Receiver sensitivity (OMA_{outer}), each lane (max) for 800GBASE-FR4 and 800GBASE-LR4

183.7.3 800GBASE-FR4 and 800GBASE-LR4 illustrative link power budgets

Illustrative power budgets and penalties for the channels are shown in Table 183-8.

Table 183-8—800GBASE-FR4 and 800GBASE-LR4 illustrative link power budgets

Parameter	800GBASE-FR4	800GBASE-LR4	Unit
Power budget (for maximum TDECQ)	7.9	11.3	dB
Operating distance	2	10	km
Channel insertion loss ^a	4 ^b	6.3 ^c	dB
Maximum discrete reflectance	-35	-35	dB
Allocation for penalties ^d (for maximum TDECQ)	3.9 ^e	5 ^f	dB

^a The channel insertion loss is calculated using the maximum distance specified in Table 183-5 and cabled optical fiber attenuation of 0.5 dB/km plus an allocation for connection and splice loss given in 183.8.2.

^b The channel insertion loss of 4 dB shown in this illustrative link power budget is valid for a maximum MPI penalty of 0.3 dB, which depends on the number and value of discrete reflectances present in the link. This channel insertion loss may be reduced by up to 0.5 dB depending on the discrete reflectances in the link as shown in Table 183-11.

^c The channel insertion loss of 6.3 dB shown in this illustrative link power budget is valid for a maximum MPI penalty of 0.4 dB, which depends on the number and value of discrete reflectances present in the link. This channel insertion loss may be reduced by up to 0.2 dB depending on the discrete reflectances in the link as shown in Table 183-12.

^d Link penalties are used for link budget calculations. They are not requirements and are not meant to be tested.

^e In this illustrative link power budget the value shown includes an allocation of 0.3 dB for MPI penalty and 0.2 dB for DGD penalty. For cases with a channel insertion loss less than 4 dB, as shown in Table 183-11, the allocation for penalties should be “7.9 – channel insertion loss”.

^f In this illustrative link power budget the value shown includes an allocation of 0.4 dB for MPI penalty and 0.7 dB for DGD penalty. For cases with a channel insertion loss less than 6.3 dB, as shown in Table 183–12, the allocation for penalties should be “11.3 – channel insertion loss”.

The values of transmitter $\text{OMA}_{\text{outer}}$ (max), transmitter $\text{OMA}_{\text{outer}}$ (min) versus $\max(\text{TECQ}, \text{TDECQ})$, and receiver sensitivity ($\text{OMA}_{\text{outer}}$) (max) versus TECQ are illustrated in Figure 183–5 for 800GBASE-FR4 and in Figure 183–6 for 800GBASE-LR4.

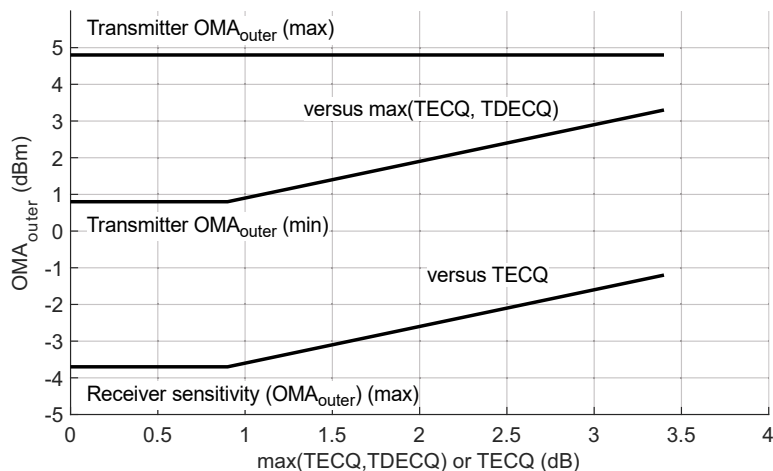


Figure 183–5—Transmitter $\text{OMA}_{\text{outer}}$ each lane versus $\max(\text{TECQ}, \text{TDECQ})$ and receiver sensitivity ($\text{OMA}_{\text{outer}}$) each lane versus TECQ for 800GBASE-FR4

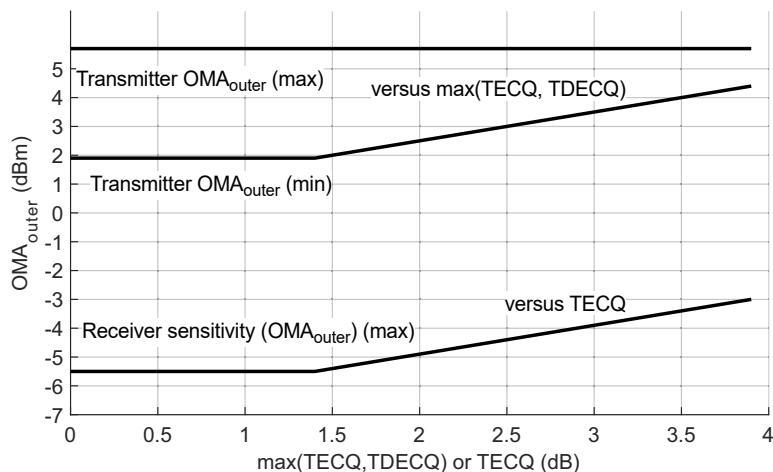


Figure 183-6—Transmitter OMA_{outer} each lane versus max(TECQ, TDECQ) and receiver sensitivity (OMA_{outer}) each lane versus TECQ for 800GBASE-LR4

183.8 Optical channel characteristics

The fiber optic cabling model is shown in Figure 183-7.

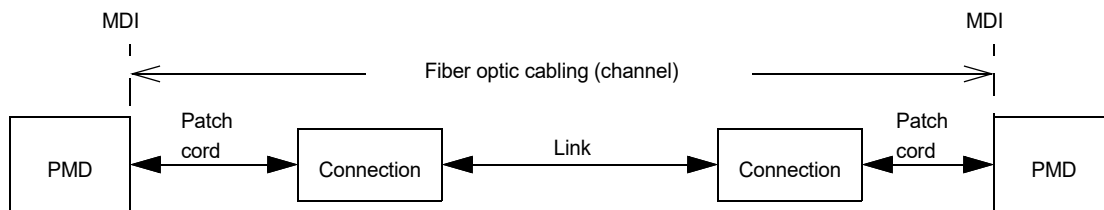


Figure 183-7—Fiber optic cabling model

The optical channel characteristics shall meet the specifications defined in Table 183-9. The fiber optic cabling consists of one or more sections of fiber optic cable and any intermediate connections required to connect sections together. A channel may contain additional connectors as long as the optical characteristics of the channel, such as attenuation, dispersion, reflections, and polarization mode dispersion meet the specifications. Insertion loss measurements of installed fiber cables are made in accordance with IEC 61280-4 one-cord reference method. The fiber optic cabling model (channel) defined here applies to each simplex fiber that makes up the duplex fiber link segment. The term channel is used here for consistency with generic cabling standards.

Table 183–9—Optical channel characteristics

Description	800GBASE-FR4	800GBASE-LR4	Unit
Operating distance (max)	2	10	km
Channel insertion loss ^{a, b} (max)	See Table 183–11	See Table 183–12	dB
Channel insertion loss (min)	0		dB
Positive dispersion ^b (max)	6.02	2.8	ps/nm
Negative dispersion ^b (min)	–11.26	–24.6	ps/nm
DGD_max ^c	2.3	4	ps
Optical return loss (min)	25	22	dB

^a These channel insertion loss values include cable, connectors, and splices.

^b Over the wavelength range 1264.5 nm to 1337.5 nm for 800GBASE-FR4, and 1294.53 nm to 1310.19 nm for 800GBASE-LR4. The dispersion specifications are based on the statistical link design methodology documented in ITU-T REC G.652, Appendix I.

^c Differential Group Delay (DGD) is the time difference at reception between the fractions of a pulse that were transmitted in the two principal states of polarization of an optical signal. DGD_max is the maximum differential group delay that the system is required to tolerate.

183.8.1 Optical fiber cable

The optical fiber cable requirements are satisfied by cables meeting the characteristics in Table 183–10. The use of optical fiber cables containing ITU-T type G.652.D (low water peak, dispersion unshifted), type G.657.A1, or type G.657.A2 (bend insensitive) fibers is recommended.

Table 183–10—Optical fiber and cable characteristics

Description	Value	Unit
Nominal fiber specification wavelength	1310	nm
Cabled optical fiber attenuation (max)	0.5 ^a	dB/km
Zero dispersion wavelength (λ_0)	$1300 \leq \lambda_0 \leq 1324$	nm
Dispersion slope (max) (S_0)	0.092	ps/(nm ² x km)

^a The 0.5 dB/km attenuation is provided for Outside Plant cable as defined in TIA 568-C.3.

183.8.2 Optical fiber connection

An optical fiber connection, as shown in Figure 183–7, consists of a mated pair of optical connectors.

The maximum link distance for 800GBASE-FR4 is based on an allocation of 1 dB total fiber cable attenuation, with the remainder of the maximum channel insertion loss specified in Table 183–11 being

allocated to connection and splice loss. For example, with no discrete reflections greater than -55 dB, a total connection and splice loss allocation of 3 dB supports six connections with an average insertion loss per connection of 0.5 dB.

The maximum link distance for 800GBASE-LR4 is based on an allocation of 5 dB total fiber cable attenuation, with the remainder of the maximum channel insertion loss specified in Table 183–12 being allocated to connection and splice loss. For example, with no discrete reflections greater than -55 dB, a total connection and splice loss allocation of 1.3 dB supports 2 connections with an average insertion loss per connection of 0.5 dB.

Connections with different loss characteristics may be used provided the requirements of Table 183–9 are met.

Table 183–11—Maximum channel insertion loss versus number of discrete reflectances for 800GBASE-FR4

Maximum channel insertion loss (dB)		Number of discrete reflectances > -55 dB and ≤ -45 dB								
		0	1	2	3	4	5	6	7	8
Number of discrete reflectance > -45 dB and ≤ -35 dB	0 to 3	4	4	4	4	4	4	4	4	4
	4	4	4	4	4	4	3.9	3.9	3.9	3.9
	5	3.9	3.9	3.9	3.9	3.9	3.8	3.8	3.8	3.8
	6	3.8	3.8	3.8	3.8	3.7	3.7	3.7	3.7	— ^a
	7	3.5	3.5	3.5	— ^a	— ^a	— ^a	— ^a	— ^a	— ^a

^a The indicated combination of reflectances does not provide a supported maximum channel insertion loss.

Table 183–12—Maximum channel insertion loss versus number of discrete reflectances for 800GBASE-LR4

Maximum channel insertion loss (dB)		Number of discrete reflectances > -55 dB and ≤ -45 dB								
		0	1	2	3	4	5	6	7	8
Number of discrete reflectances > -45 dB and ≤ -35 dB	0 to 5	6.3	6.3	6.3	6.3	6.3	6.3	6.3	6.3	6.3
	6	6.3	6.3	6.3	6.3	6.2	6.2	6.2	6.2	6.2
	7	6.2	6.2	6.2	6.2	6.1	6.1	6.1	6.1	— ^a
	8	6.1	6.1	6.1	— ^a	— ^a	— ^a	— ^a	— ^a	— ^a

^a The indicated combination of reflectances does not provide a supported maximum channel insertion loss.

183.8.3 Medium Dependent Interface (MDI)

The 800GBASE-FR4 or 800GBASE-LR4 PMD is coupled to the fiber optic cabling at the MDI. The MDI is the interface between the PMD and the fiber optic cabling (as shown in Figure 183–7). Examples of an MDI include the following:

- Connectorized fiber pigtail
- PMD receptacle

When the MDI is a connector plug and receptacle connection, it shall meet the interface performance specifications of IEC 61753-1 and IEC 61753-021-02.

NOTE—Transmitter compliance testing is performed at TP2 as defined in 183.5.1, not at the MDI.

183.9 Definition of optical parameters and measurement methods

All transmitter optical measurements shall be made through a short patch cable, between 2 m and 5 m in length, unless otherwise specified.

183.9.1 Test patterns for optical parameters

While compliance is to be achieved in normal operation, specific test patterns are defined for measurement consistency and to enable measurement of some parameters. Table 183–14 gives the test patterns to be used in each measurement, unless otherwise specified, and also lists references to the subclauses in which each parameter is defined. Any of the test patterns given for a particular test in Table 183–14 may be used to perform that test. The test patterns used in this clause are shown in Table 183–13.

Table 183–13—Test patterns

Pattern	Pattern description	Defined in
3	PRBS31Q	177.6.1.4
4	PRBS13Q	177.6.1.3
5	Scrambled idle test pattern encoded by the 800GBASE-R Inner FEC	175.2.4.11, 177.4
6	SSPRQ	177.6.1.5
7	Valid 800GBASE-R signal encoded by the 800GBASE-R Inner FEC	175.2.4, 177.4
8	PRBS31 pattern encoded by 800GBASE-R Inner FEC	177.6.1.1

Table 183–14—Mapping of parameters to test patterns and related subclauses

Parameter	Pattern	Related subclause
Wavelength	3, 4, 5, 6, 7 or 8	183.9.3
Side mode suppression ratio	3, 5, 6, 7 or 8	183.9.3
Average optical power	3, 5, 6, 7 or 8	183.9.4
Outer optical modulation amplitude (OMA _{outer})	4 or 6	183.9.5
Transmitter and dispersion eye closure for PAM4 (TDECQ)	6	183.9.6
Transmitter eye closure for PAM4 (TECQ)	6	183.9.7
Transmitter functional test	8	183.9.8
Overshoot and undershoot	6	183.9.9
Transmitter power excursion	6	183.9.10
Extinction ratio	4 or 6	183.9.11
Transmitter transition time	6	183.9.12
RIN _x OMA	4 or 6	183.9.13
Receiver sensitivity	8	183.9.14
Stressed receiver conformance test signal calibration	6	183.9.15
Stressed receiver sensitivity	8	183.9.15

183.9.2 Reference receiver

A reference receiver is defined for transmitter measurements. The reference receiver is composed of an O/E converter, a clock recovery unit (CRU), and a pattern-triggered oscilloscope as shown in Figure 180–7. The combined frequency response of the O/E converter and oscilloscope has a 3 dB bandwidth of approximately 56.71875 GHz with a fourth-order Bessel-Thomson response for frequencies up to at least 1.3 x 113.4375 GHz and a response no higher than –20 dB for frequencies greater than 1.3 x 113.4375 GHz.

The CRU has a corner frequency of 4.27 MHz and a slope of 20 dB/decade. The CRU may be implemented in hardware or software depending on oscilloscope technology.

183.9.3 Wavelength and side mode suppression ratio (SMSR)

The wavelength and SMSR of each optical lane shall be within the ranges given in Table 183–6 if measured per IEC 61280-1-3. The lane under test is modulated using one of the test patterns defined in Table 183–14.

183.9.4 Average optical power

The average optical power of each lane shall be within the limits given in Table 183–6 if measured using the methods given in IEC 61280-1-1. The average optical power is measured using one of the test patterns defined in Table 183–14.

183.9.5 Outer optical modulation amplitude (OMA_{outer})

The OMA_{outer} of each lane shall be within the limits given in Table 183–6. OMA_{outer} is measured using a test pattern specified for OMA_{outer} in Table 183–14 as the difference between the average optical launch power level P_3 , measured over the central 2 UI of a run of 7 threes, and the average optical launch power level P_0 , measured over the central 2 UI of a run of 6 zeros, as shown in Figure 183–8. OMA_{outer} is measured using the waveforms captured at the output of the reference receiver defined in 183.9.2.

When measured with pattern 6, OMA_{outer} is measured as the difference between the average optical launch power level P_3 , measured over the central 2 UI of the first 7 UI of all runs of 7 or more threes, and the average optical launch power level P_0 , measured over the central 2 UI of the first 6 UI of all runs of 6 or more zeros.

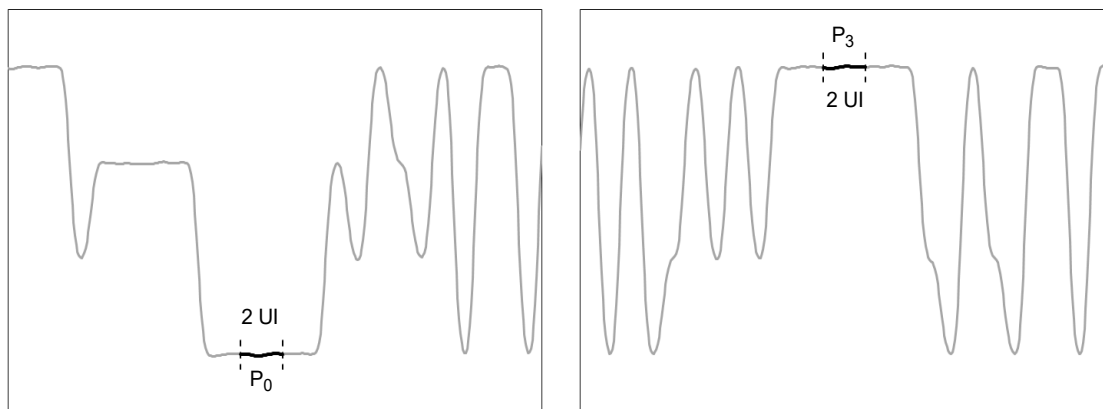


Figure 183–8—Example power levels P_0 and P_3 from PRBS13Q test pattern

183.9.6 Transmitter and dispersion eye closure for PAM4 (TDECQ)

The TDECQ of each lane shall be within the limit given in Table 183–6. The TDECQ is measured using the methods specified in 180.9.6, with the following exceptions:

- The signaling rate of the test pattern generator is as given in Table 183–6 and uses the test pattern specified for TDECQ in Table 183–14.
- The reference receiver is as specified in 183.9.2
- The target PAM4 symbol error ratio is 9.6×10^{-3} and the related Q_t value is 2.489.
- The normalized noise power density spectrum, $N(f)$ is equivalent to white noise filtered by a fourth-order Bessel-Thomson response filter with a 3 dB bandwidth of 56.71875 GHz
- The clock recovery unit (CRU) has a corner frequency of 4.27 MHz.
- The optical return loss is as given in Table 183–6.
- The aggressor lanes shall have power levels as specified in Table 183–7 for the stressed receiver sensitivity test.
- For those cases where there is an xAUI-n chip-to-chip (C2C) or chip-to-module (C2M) interface (see Table 183–1), the clock source for the SSPRQ test pattern is derived from the clock recovered from

- the xAUI-n input signal. The pattern of the xAUI-n input signal may be PRBS31Q, scrambled idle, or a valid xBASE-R signal.
- The transmitter is tested using a worst case optical channel specified in 183.9.6.1.
- The conformance test setup uses the same block diagram as shown in Figure 180–9 with the exception that a demultiplexing filter is used in front of the reference receiver to separate the optical signal of the lane under test.

183.9.6.1 Channel requirements

The transmitter is tested using an optical channel that meets the requirements listed in Table 183–15.

Table 183–15—Transmitter compliance channel specifications

PMD type	Lane	Dispersion ^a (ps/nm)		Insertion loss ^b	Optical return loss ^c (dB)	Max mean DGD ^d (ps)
		Minimum	Maximum			
800GBASE-FR4	0	$0.203 \times (\lambda - 1271) - 9.96$	$0.187 \times (\lambda - 1271) - 5.81$	Minimum	17.1	0.8
	1	$0.194 \times (\lambda - 1291) - 5.99$	$0.183 \times (\lambda - 1291) - 2.12$			
	2	$0.185 \times (\lambda - 1311) - 2.22$	$0.177 \times (\lambda - 1311) + 1.47$			
	3	$0.176 \times (\lambda - 1331) + 1.38$	$0.169 \times (\lambda - 1331) + 4.92$			
800GBASE-LR4	All	$0.225 \times \lambda \times [1 - (1321.1 / \lambda)^4]$	$0.2175 \times \lambda \times [1 - (1307 / \lambda)^4]$	Minimum	15.6	0.8

^a The dispersion is measured for the wavelength of the transmitter lane under test (λ in nm). The coefficient assumes 2 km for 800GBASE-FR4 and 10 km for 800GBASE-LR4. The dispersion specifications are based on the statistical link design methodology documented in ITU-T REC G.652, Appendix I.

^b The value “Minimum” implies that the test channel insertion loss does not excessively degrade the signal to noise ratio of the measurement.

^c The back-reflection is applied at TP2.

^d The limit for maximum mean DGD is to ensure that the contribution to TDECQ from DGD is substantially less than the primary contribution due to chromatic dispersion.

A 800GBASE-FR4 or 800GBASE-LR4 transmitter is to be compliant with a total dispersion at least as negative as the “minimum dispersion” and at least as positive as the “maximum dispersion” columns specified in Table 183–15 for the wavelength of the transmitter lane under test. This may be achieved with channels consisting of fibers with lengths chosen to meet the dispersion requirements.

To verify that the fiber has the correct amount of dispersion, the measurement method defined in IEC 60793-1-42 may be used. The measurement is made in the linear power regime of the fiber.

The channel provides back reflection with return loss specified in Table 183–15. The state of polarization of the back reflection is adjusted to create the greatest RIN.

The mean DGD of the channel is to be less than the value specified in Table 183–15.

183.9.7 Transmitter eye closure for PAM4 (TECQ)

The transmitter eye closure for PAM4 (TECQ) is a measure of the optical transmitter's eye closure at TP2. The TECQ of each lane shall be within the limit given in Table 183–6 if measured using a test pattern specified for TECQ in Table 183–14. The TECQ of each lane is measured using the methods specified for TDECQ in 183.9.6, except that the test fiber is not used.

183.9.8 Transmitter functional test

The transmitter functional symbol error histogram is measured using the method defined in 180.9.8 with the following exceptions:

- The transmitter under test is connected to the functional receiver (FnRx) by a test fiber which meets the requirements in 183.9.6.1. Full compliance with 183.9.6.1 requires the test fiber to provide back reflection with polarization state resulting in the greatest RIN. Other test fibers, including patch cords, may be used for additional tests.

where in 180.9.8.1

- The FnRx is a variable optical attenuator (VOA) followed by an optical receiver (ORx) that complies with characteristics as given in Table 183–7. The VOA and test fiber approximate the compliance channel as given in Table 183–15 with the exception that the channel loss is not minimum.

where in Equation (180–16)

- $Tx_DUT_power_budget$ is the transmitter under test power budget for the specific fiber under test as given in Table 183–8, except uses measured instead of max TDECQ value and is given by Equation (180–17).

where in Equation (180–17)

- $Channel_insertion_loss$ is the “Channel insertion loss” as given in Table 183–8.
- $MPI_and_DGD_penalties_allocation$ is the allocation for MPI and DGD penalties as given in Table 183–8.

where in 180.9.8.2

- The transmitter functional symbol error histogram mask per each lane is given in Table 180–17 and is measured using the test pattern as given in Table 183–14.

183.9.9 Transmitter overshoot and undershoot

The transmitter overshoot and undershoot of each lane shall be within the limit given in Table 183–6 if measured using a test pattern specified for overshoot and undershoot in Table 183–14.

Transmitter overshoot and undershoot are measured using the waveform captured for the TDECQ test (see 183.9.6) and the waveform captured for the TECQ test (see 183.9.7), at the output of the reference receiver defined in 183.9.2, before the reference equalizer.

Transmitter overshoot and undershoot are calculated using the methods in 140.7.7, with the exceptions that P_0 , P_3 and OMA_{outer} are measured using the methods in 183.9.5 and the hit ratio used to determine P_{max} and P_{min} is 1×10^{-4} .

183.9.10 Transmitter power excursion

The transmitter power excursion of each lane shall be within the limit given in Table 183–6 if measured using a test pattern specified for transmitter power excursion in Table 183–14.

Transmitter power excursion is measured using waveforms captured at the output of the reference receiver defined in 183.9.2.

Transmitter power excursion is defined in 140.7.8, with the exception that the hit ratio used to determine $P_{\max}$ and $P_{\min}$ is 1×10^{-4} .

183.9.11 Extinction ratio

The extinction ratio of each lane shall be within the limit given in Table 183–6 if measured using a test pattern specified for extinction ratio in Table 183–14. The extinction ratio of a PAM4 optical signal is defined as the ratio of the average optical launch power levels P_3 and P_0 , as defined for OMA_{outer} (see 183.9.5).

183.9.12 Transmitter transition time

The transmitter transition time of each lane shall be within the limit given in Table 183–6 if measured using a test pattern specified for transmitter transition time in Table 183–14.

Transmitter transition time is defined as the slower of the time interval of the transition from 20% of OMA_{outer} to 80% of OMA_{outer} or from 80% of OMA_{outer} to 20% of OMA_{outer} , for the rising and falling edges respectively. Transmitter transition time is measured using waveforms captured at the output of the reference receiver defined in 183.9.2.

The 0% level and the 100% level are P_0 and P_3 as defined by the OMA_{outer} measurement procedure (see 183.9.5). When the SSPRQ pattern is used, the rising edge used for the measurement is that within the 00000333333 symbol sequence and the falling edge is that within the 33333000000 symbol sequence.

183.9.13 Relative intensity noise (RIN_xOMA)

RIN_xOMA , with “x” referring to the value for optical return loss tolerance in Table 183–6, shall be within the limit given in Table 183–6 when measured using the test pattern and sampling range specified for OMA_{outer} measurement in 183.9.5, but with applied x dB optical reflection and the reference receiver specified in 183.9.2. RIN_xOMA is measured using the methods specified in 180.9.13.

183.9.14 Receiver sensitivity

The receiver sensitivity (OMA_{outer}) of each lane shall be within the limits given in Table 183–7 if measured using a test pattern specified for receiver sensitivity in Table 183–14.

The conformance test signal applied at TP3 meets the requirements for a 800GBASE-FR4 or 800GBASE-LR4 transmitter followed by an attenuator. An optical demultiplexer may be used to separate the lane having the wavelength for the lane under test as specified in Table 183–7 for calibrating the TECQ.

The TECQ of the conformance test signal is measured according to 183.9.7, except that the test fiber is not used. The measured value of TECQ is then used to calculate the limit for receiver sensitivity (OMA_{outer}) as specified in Table 183–7.

Precoding (see 176.7.1.2) is enabled if the receiver requests precoding using the ILT function.

The measured receiver sensitivity is the lowest value of OMA_{outer} where the PMD receiver meets the BLER requirement in 183.2, measured at the PMA using the test method in either 174A.9.5 or 174A.9.6 with parameters provided in Table 182–16 and $p = 4$. The error mask $H_{\max}(k)$ to be used in the method of 174A.9.5 is provided in Table 180–20 in the column for $p = 4$. When testing using 174A.9.5 or 174A.9.6, the compliance shall not be further verified using 174A.9.7.

The Low-stressed Receiver Sensitivity (LRS) conformance test signal at TP3 meets the transmit characteristics given in Table 183–6 except for characteristics in Table 180–18.

NOTE—In order to predict whether a receiver meets the BLER requirement in a short test time, extrapolation of the measured histogram (see 174A.9.3) to $H_m(16)$ can be performed. One way of doing this is using a linear fit of $\log_{10}(H_m(k))$ for $k = (n - 2)$ to n , where these values of k are the three highest bins having a count of 10 or more. If the slope of the histogram over the measured bins is increasing, it might be an indication of not meeting the BLER requirement. To verify that the slope is not increasing, the values of $(\log_{10}(H_m(k)) - \log_{10}(H_m(4))) / (k - 4)$ should be less than $(\log_{10}(H_m(4)) - \log_{10}(H_m(2))) / 2$ for each k greater than 4.

183.9.15 Stressed receiver sensitivity

Stressed receiver sensitivity shall be within the limit given in Table 183–7 if measured using the method defined in 122.8.10, with the following exceptions:

- The SECQ of the stressed receiver conformance test signal is measured according to 183.9.6, except that the test fiber is not used. The transition time of the stressed receiver conformance test signal is no greater than the value specified in Table 183–6.
- With the Gaussian noise generator on and the sinusoidal jitter and sinusoidal interferer turned off, the RIN_x OMA of the SRS test source (where x is the value for optical return loss tolerance from Table 183–6) shall be no greater than the value specified in Table 183–6.
- An example stressed receiver conformance test setup is shown in Figure 122–8; however, alternative test setups that generate equivalent stress conditions may be used.
- The frequency of the sinusoidal interferer is greater than 200 MHz.
- The signaling rate of the test pattern generator and the extinction ratio of the E/O converter are as given in Table 183–6.
- The values of transmitter overshoot and undershoot and transmitter power excursion of the stressed receiver conformance test signal are within the limits specified in Table 183–6.
- The stressed receiver conformance input signal meets the sinusoidal jitter specified in 183.9.15.1.
- The required values of the “Stressed receiver sensitivity (OMA_{outer}), each lane (max)”, “Stressed eye closure for PAM4 (SECQ), lane under test” and “OMA_{outer} of each aggressor lane” are as given in Table 183–7.

The measured receiver sensitivity is the lowest value of OMA_{outer} where the receiver meets the requirements in 183.2 using the test method in either 174A.9.5 or 174A.9.6 with parameters provided in Table 182–16 and $p = 4$. The error mask $H_{\max}(k)$ to be used in the method of 174A.9.5 is provided in Table 180–20 in the column for $p = 4$. When testing using 174A.9.5 or 174A.9.6, the compliance shall not be further verified using 174A.9.7.

Precoding (see 176.7.1.2) is enabled if the receiver requests precoding using the ILT function.

NOTE—In order to predict whether a receiver meets the BLER requirement in a short test time, extrapolation of the measured histogram (see 174A.9.3) to $H_m(16)$ can be performed. One way of doing this is using a linear fit of $\log_{10}(H_m(k))$ for $k = (n - 2)$ to n , where these values of k are the three highest bins having a count of 10 or more. If the slope of the histogram over the measured bins is increasing, it might be an indication of not meeting the BLER requirement. To verify that the slope is not increasing, the values of $(\log_{10}(H_m(k)) - \log_{10}(H_m(4))) / (k - 4)$ should be less than $(\log_{10}(H_m(4)) - \log_{10}(H_m(2))) / 2$ for each k greater than 4.

183.9.15.1 Sinusoidal jitter for receiver conformance test

The sinusoidal jitter is used to test receiver jitter tolerance. The amplitude of the applied sinusoidal jitter is dependent on frequency as specified in Table 182–17 and is illustrated in Figure 182–8.

183.10 Safety, installation, environment, and labeling

183.10.1 General safety

Equipment subject to this clause shall conform to the general safety requirements in J.2.

183.10.2 Laser safety

Optical transceivers shall conform to Hazard Level 1 laser requirements as defined in IEC 60825-1 and IEC 60825-2, under any condition of operation. This includes single fault conditions whether coupled into a fiber or out of an open bore.

Conformance to additional laser safety standards may be required for operation within specific geographic regions.

Laser safety standards and regulations require that the manufacturer of a laser product provide information about the product's laser, safety features, labeling, use, maintenance, and service. This documentation explicitly defines requirements and usage restrictions on the host system necessary to meet these safety certifications.¹⁵

183.10.3 Installation

It is recommended that proper installation practices, as defined by applicable local codes and regulation, be followed in every instance in which such practices are applicable.

183.10.4 Environment

Normative specifications in this clause shall be met by a system integrating a 800GBASE-FR4 or 800GBASE-LR4 PMD over the life of the product while the product operates within the manufacturer's range of environmental, power, and other specifications.

It is recommended that manufacturers indicate in the literature associated with the PHY the operating environmental conditions to facilitate selection, installation, and maintenance.

It is recommended that manufacturers indicate, in the literature associated with the components of the optical link, the distance and operating environmental conditions over which the specifications of this clause are met.

183.10.5 Electromagnetic emission

A system integrating a 800GBASE-FR4 or 800GBASE-LR4 PMD shall comply with applicable local and national codes for the limitation of electromagnetic interference.

183.10.6 Temperature, humidity, and handling

The optical link is expected to operate over a reasonable range of environmental conditions related to temperature, humidity, and physical handling (such as shock and vibration). Specific requirements and values for these parameters are considered to be beyond the scope of this standard.

¹⁵A host system that fails to meet the manufacturer's requirements and/or usage restrictions may emit laser radiation in excess of the safety limits of one or more safety standards. In such a case, the host manufacturer is required to obtain its own laser safety certification.

183.10.7 PMD labeling requirements

It is recommended that each PHY (and supporting documentation) be labeled in a manner visible to the user, with at least the applicable safety warnings and the applicable port type designation (e.g., 800GBASE-FR4).

Labeling requirements for Hazard Level 1 lasers are given in the laser safety standards referenced in 183.10.2.

183.11 Management variables

PMD control and status variables intended to be accessible via a management system are listed in Table 183–16 and Table 183–17. Additional variables associated with the APSU functions are listed in Table 178B–3 and Table 178B–4.

If the MDIO interface (see Clause 45) is implemented, PMD control and status variables shall be mapped to MDIO registers as shown in Table 183–16 and Table 183–17.

If the MDIO interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 183–16—MDIO/PMD control variable mapping

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_reset	183.5.6	1.0.15	45.2.1.242.1
Global_PMD_transmit_disable	183.5.7	1.9.0	45.2.1.8
PMD_transmit_disable _{n-1} to PMD_transmit_disable ₀	183.5.7	1.9.4 to 1.9.1	45.2.1.8

Table 183–17—MDIO/PMD status variable mapping

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_fault	183.5.9	1.1.7	45.2.1.2.3
PMD_transmit_fault	183.5.10	1.8.11	45.2.1.7
PMD_receive_fault	183.5.11	1.8.10	45.2.1.7
Global_PMD_signal_detect	183.5.4	1.10.0	45.2.1.9
PMD_signal_detect _{n-1} to PMD_signal_detect ₀	183.5.5	1.10.4 to 1.10.1	45.2.1.9

183.12 Protocol implementation conformance statement (PICS) proforma for Clause 183, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-FR4 and 800GBASE-LR4¹⁶

183.12.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 183, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-FR4 and 800GBASE-LR4, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

183.12.2 Identification

183.12.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

183.12.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3-202x, Clause 183, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-FR4 and 800GBASE-LR4
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3-202x.)	

Date of Statement	
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¹⁶*Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

183.12.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*FR4	800GBASE-FR4	183.7	Device supports requirements for 800GBASE-FR4 PHY	O.1	Yes [] No []
*LR4	800GBASE-LR4	183.7	Device supports requirements for 800GBASE-LR4 PHY	O.1	Yes [] No []
*INS	Installation / cable	183.8	Items marked with INS include installation practices and cable specifications not applicable to a PHY manufacturer	O	Yes [] No []
DC	Delay constraints	183.4.1	Device conforms to delay constraints	M	Yes []
SC	Skew constraints	183.4.2	Device conforms to Skew and Skew Variation constraints	M	Yes []
*MD	MDIO capability	183.11	Registers and interface supported	O	Yes [] No []

183.12.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-FR4 and 800GBASE-LR4

183.12.4.1 PMD functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
F1	Compatible with 800GBASE-R PCS and PMA	183.1		M	Yes []
F2	Block error ratio	183.2	Meets the BLER specified in 183.2	M	Yes []
F3	Transmit function	183.5.2	Conveys symbols from PMD service interface to MDI	M	Yes []
F4	Optical multiplexing and delivery to the MDI	183.5.2	Optically multiplexes the four optical signals for delivery to the MDI	M	Yes []
F5	Mapping between optical signal and logical signal for transmitter	183.5.2	Optical power levels from lowest to highest correspond to tx_symbols zero, one, two, and three, respectively	M	Yes []
F6	Receive function	183.5.3	Conveys symbols from MDI to PMD service interface	M	Yes []
F7	Conversion of four optical signals to symbol streams	183.5.3	For delivery to the PMD service interface	M	Yes []
F8	Mapping between optical signal and logical signal for receiver	183.5.3	Optical power levels from lowest to highest correspond to rx_symbols zero, one, two, and three, respectively	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
F9	Global Signal Detect behavior	183.5.4	Global_PMD_signal_detect is an indicator of the presence of an optical signal on all lanes	M	Yes []
F10	Lane-by-lane Signal Detect function	183.5.5	Sets PMD_signal_detect_i values on a lane-by-lane basis per requirements of Table 183–2	M	Yes []
F11	PMD reset function	183.5.6	Resets the PMD sublayer	O	Yes [] No []
F12	Global transmit disable function	183.5.7	Disables all of the optical transmitters with the Global_PMD_transmit_disable variable	O	Yes [] No []
F13	PMD lane-by-lane transmit disable using management variable	183.5.8	Disables the optical transmitter on the lane associated with the PMD_transmit_disable_i variable	O	Yes [] No []
F14	PMD lane-by-lane transmit disable using state variable	183.5.8	Disables the optical transmitter when tx_disable variable is true	M	Yes []
F15	PMD_fault function	183.5.9	Sets PMD_fault to one if any local fault is detected	O	Yes [] No []
F16	PMD_transmit_fault function	183.5.10	Sets PMD_transmit_fault to one if a local fault is detected on any transmit lane	O	Yes [] No []
F17	PMD_receive_fault function	183.5.11	Sets PMD_receive_fault to one if a local fault is detected on any receive lane	O	Yes [] No []

183.12.4.2 PMD to MDI optical specifications for 800GBASE-FR4

Item	Feature	Subclause	Value/Comment	Status	Support
FRF1	Transmitter meets specifications in Table 183–6	183.7.1	Per definitions in 183.9	FR4:M	Yes [] N/A []
FRF2	Receiver meets specifications in Table 183–7	183.7.2	Per definitions in 183.9	FR4:M	Yes [] N/A []

183.12.4.3 PMD to MDI optical specifications for 800GBASE-LR4

Item	Feature	Subclause	Value/Comment	Status	Support
FRF1	Transmitter meets specifications in Table 183–6	183.7.1	Per definitions in 183.9	LR4:M	Yes [] N/A []
FRF2	Receiver meets specifications in Table 183–7	183.7.2	Per definitions in 183.9	LR4:M	Yes [] N/A []

183.12.4.4 Optical measurement methods

Item	Feature	Subclause	Value/Comment	Status	Support
OM1	Measurement cable	183.9	2 m to 5 m in length	M	Yes []
OM2	Center wavelength and SMSR	183.9.3	Per IEC 61280-1-3 under modulated conditions	M	Yes []
OM3	Average optical power	183.9.4	Per IEC 61280-1-1	M	Yes []
OM4	OMA measurements	183.9.5	Each lane	M	Yes []
OM5	Transmitter and dispersion eye closure for PAM4 (TDECQ)	183.9.6	Each lane	M	Yes []
OM6	Transmitter eye closure for PAM4 (TECQ)	183.9.7	Each lane	M	Yes []
OM8	Transmitter functional symbol error histogram	183.9.8	Each lane	M	Yes []
OM8	Transmitter overshoot and undershoot	183.9.9	Each lane	M	Yes []
OM9	Transmitter power excursion	183.9.10	Each lane	M	Yes []
OM10	Extinction ratio	183.9.11	Each lane	M	Yes []
OM11	Transmitter transition time	183.9.12	Each lane	M	Yes []
OM12	Relative intensity noise	183.9.13	Each lane	M	Yes []
OM13	Receiver sensitivity	183.9.14	Each lane	M	Yes []
OM14	Stressed receiver sensitivity	183.9.15	Each lane	M	Yes []

183.12.4.5 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
ES1	General safety	183.10.1	Conforms to J.2	M	Yes []
ES2	Laser safety—IEC Hazard Level 1	183.10.2	Conforms to Hazard Level 1 laser requirements defined in IEC 60825-1 and IEC 60825-2	M	Yes []
ES3	Electromagnetic interference	183.10.5	Complies with applicable local and national codes for the limitation of electromagnetic interference	M	Yes []

183.12.4.6 Characteristics of the fiber optic cabling and MDI

Item	Feature	Subclause	Value/Comment	Status	Support
OC1	Fiber optic cabling	183.8	Meets requirements specified in Table 183–9 and Table 183–10	INS:M	Yes [] N/A []
OC2	Maximum discrete reflectance	183.8.2	Meets requirements specified in Table 183–10	INS:M	Yes [] N/A []
OC3	MDI Requirements	183.8.3	Meets IEC 61753-1 and IEC 61753-021-2	INS:M	Yes [] N/A []

183.12.4.7 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Alternate access to management variables is provided	183.11		!MD:M	Yes [] N/A []
M2	Mapping of MDIO control bits and MDIO status bits	183.11		MD:M	Yes [] N/A []